COURSE 2 · SECTION 11 · WORKED SOLUTION

Capstone: the valve seat ring story

Section 11 · Capstone dashboard

DESIGN QUESTION

Can one report page tell the P300 valve seat ring story from drawing review to field ppm, and let a reviewer drill into any drawing set?

Steps

1. Open Section11_Start.pbix > File > Save as > Engxtech_S11. Add page 'Seat ring story' (Canvas 16:9).
2. Copy and paste the six visuals from the Findings, Stacks, FEA & Cpk, Changes and Validation & field pages, and change them as in the brief's visual table.
3. Slicers: Parts[PartID] (Dropdown), SeatRingRevs[SeatRingRev] (Tile), BuildMonths[BuildMonth] (Between). Format > Edit interactions: Part slicer filters visuals 1, 3, 4 only; the SeatRingRev and BuildMonth slicers filter visual 6 only.
4. Page 'Drawing detail': Build visual > Drill through well: Drawings[PartRev]; Keep all filters On. Tables for Drawings and ReviewFindings, cards Finding Count and Major Findings, and the category Pareto copied from the Findings page.
5. Page 'ppm counts': Allow use as tooltip On; Canvas Type Tooltip. Cards Wear Failures 90d, Matured Units, Field ppm, CI Low ppm, CI High ppm. On visual 6: Format > General > Tooltips > Type Report page > Page ppm counts.
6. View > Bookmarks: clear the slicers > Add 'Whole portfolio'; Part = P300 > Add 'Seat ring focus'. Insert > Buttons > Blank, Action Bookmark, one per bookmark.
7. Right-click the P300-B bar in visual 1 > Drill through > Drawing detail. Compare with the table below.

No new DAX: every measure comes from an earlier section (all in _Measures)

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Visual 1  Finding Count (S04)                        by Drawings[PartRev], legend ReviewFindings[Category]
Visual 2  Req Min/Max, WC Min/Max/Result, RSS Min/Max/Result (S05), rows SeatRingBuild[Build], filter STK-01
Visual 3  Min SF by FEA[Case], SF Colour, constant line 1.5 (S05)
Visual 4  Min Cpk by ManufacturingCpk[MachineID], Cpk Colour, constant line 1.33 (S05)
Visual 5  TriageDays, ApprovalDays, ImplementDays, CloseDays for ECR-3200 (S08)
Visual 6  Field ppm by SeatRingRevs[SeatRingRev], error bars CI Low ppm / CI High ppm (S09)
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Why visual 1 is findings by part-rev and not the category Pareto: drill-through needs the drill field (Drawings[PartRev]) on the source visual. The category Pareto moves to the Drawing detail page, where it shows the categories of the drawing set you picked.

A sound caption for the ppm chart: 'Rev B 2,892 ppm on 81 failures. Rev C 1,710 ppm on 4 failures in one matured build month: an early signal, interval includes rev B's rate and overlaps rev A.' Avoid 'fixed', 'solved' or a percentage improvement for rev C.

Expected result

Check	Result
P300-B review findings	**9**
P300 review findings, all revs	**15**
Drill-through P300-B: drawing / days late / rounds	**DRW-1008 / 19 d / 3**
Drill-through P300-B: major findings / Tolerance	**4 / 6 of 9**
STK-01 rev C: WC / RSS (mm)	**0.006..0.094 (fail) / 0.0298..0.0702 (pass)**
P300 seating load SF, rev B / rev C	**1.53 / 1.78**
GRD-02 Cpk (P300)	**0.787**
ECR-3200 cycle / approval stage	**123 d / 57 d**
Rev B ppm (95 % CI)	**2892 ppm (81/28007); 95 % CI 2297-3595 ppm**
Rev C ppm (95 % CI)	**1710 ppm (4/2339); 95 % CI 466-4379 ppm**